

A Counterexample to the Filter- π FDD Question

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Source Question

The source paper is Tomasz Kania and Jaroslaw Swaczyna, *The π -property of a Banach space along a filter*, arXiv:2508.21242. Question 1.2 asks:

Suppose that a separable Banach space has the $\mathcal{F}$ - π -property with respect to a certain filter $\mathcal{F}$. Does X have a finite-dimensional decomposition (FDD)?

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Result

Theorem 1. *Every separable Banach space X has the $\mathcal{F}$ - π -property for some filter $\mathcal{F}$ on $\mathbb{N}$ containing the Frechet filter.*

Corollary 1. *Question 1.2 of arXiv:2508.21242 has a negative answer. Indeed, Enflo's separable Banach space without the approximation property has the $\mathcal{F}$ - π -property for some such filter $\mathcal{F}$, but it does not have an FDD.*

Proof Intuition

The filter is allowed to depend on the space and on the chosen sequence of finite-rank projections. A separable Banach space has enough finite-rank projections to approximate the identity on any fixed finite set of vectors: project onto the finite-dimensional span of those vectors. Separability lets us choose a countable strong-operator dense list of norm-bounded finite-rank projections, and we repeat every projection in that list infinitely many times. The desired filter is then generated by the sets of indices on which this sequence approximates a given vector. Finite intersections stay infinite because any finite list of vector constraints is handled by one finite-rank projection, which appears infinitely often.

Proof

Lemma 1. *Let X be separable. For each $m \in \mathbb{N}$, the set*

$$\mathcal{P}_m = \{P \in \mathcal{L}(X) : P^2 = P, \dim P[X] < \infty, \|P\| \leq m\}$$

has a countable dense subset for the strong operator topology restricted to $\mathcal{P}_m$.

Proof. Fix a dense sequence (x_j) in X . On the norm ball $\{T \in \mathcal{L}(X) : \|T\| \leq m\}$, the metric

$$d(S, T) = \sum_{j=1}^{\infty} 2^{-j} \min\{1, \|(S - T)x_j\|\}$$

induces the strong operator topology. This ball embeds into the countable product $\prod_j m\|x_j\|B_X$, which is separable because X is separable. Hence the ball is separable in this topology, and so is its subspace $\mathcal{P}_m$. $\square$

For each m , choose a countable strong-operator dense set $\mathcal{D}_m \subseteq \mathcal{P}_m$. Let (Q_n) be a sequence that enumerates $\bigcup_m \mathcal{D}_m$ and repeats each of its elements infinitely many times.

We first record the finite approximation property of this sequence. Given vectors $x_1, \dots, x_r \in X$ and $\varepsilon > 0$, let $E = \text{span}\{x_1, \dots, x_r\}$. Since E is finite dimensional, it is complemented in X : there is a bounded projection $P : X \rightarrow E$. Pick $m > \|P\|$. By strong-operator density of $\mathcal{D}_m$, there is $Q \in \mathcal{D}_m$ such that

$$\max_{1 \leq i \leq r} \|Qx_i - Px_i\| < \varepsilon.$$

Since $Px_i = x_i$, this says $\max_i \|Qx_i - x_i\| < \varepsilon$. Because the projection Q occurs infinitely many times in (Q_n) , the set of indices n satisfying this finite list of inequalities is infinite.

For $x \in X$ and $k \in \mathbb{N}$, set

$$A(x, k) = \{n \in \mathbb{N} : \|Q_n x - x\| < 1/k\}.$$

The preceding paragraph implies that every finite intersection of sets $A(x, k)$, even after intersecting with an arbitrary cofinite subset of $\mathbb{N}$, is nonempty; in fact it is infinite. Therefore the family consisting of all cofinite subsets of $\mathbb{N}$ together with all $A(x, k)$ generates a proper filter $\mathcal{F}$ containing the Frechet filter.

Now fix $x \in X$ and $\varepsilon > 0$. Choose k with $1/k < \varepsilon$. Since $A(x, k) \in \mathcal{F}$, for all $n \in A(x, k)$ we have $\|Q_n x - x\| < \varepsilon$. Thus

$$\lim_{n, \mathcal{F}} \|Q_n x - x\| = 0.$$

The sequence (Q_n) consists of finite-rank projections, so it witnesses the $\mathcal{F}$ - π -property of X .

For the corollary, let X be Enflo's separable Banach space without the approximation property. The theorem gives the $\mathcal{F}$ - π -property for some filter $\mathcal{F}$ containing the Frechet filter. If X had an FDD (F_n) , the partial-sum projections $S_N : X \rightarrow F_1 + \dots + F_N$ would be finite-rank and would converge pointwise to the identity. By the uniform boundedness principle, $\sup_N \|S_N\| < \infty$. Hence X would have the bounded approximation property, and therefore the approximation property, contradicting Enflo's construction. Thus this X is a counterexample to Question 1.2.

Finally, if the intended reading of the question is restricted to analytic filters, the same counterexample persists. Theorem B of the source paper says that whenever a sequence witnesses the $\mathcal{F}$ - π -property for an arbitrary filter $\mathcal{F}$, there is an analytic filter $\mathcal{G} \subseteq \mathcal{F}$ for which the same sequence is still a witness. If needed to maintain the paper's convention that all filters contain the Frechet filter, replace $\mathcal{G}$ by the filter generated by $\mathcal{G} \cup \mathcal{F}_{\text{Fr}}$. This filter is still analytic, is still contained in $\mathcal{F}$, contains the Frechet filter, and still contains the sets that witness convergence.

Verification Notes

The filter contains the Frechet filter by construction. The proof does not use any approximation property of X ; it uses only separability and the elementary fact that finite-dimensional subspaces

of Banach spaces are complemented. The resulting sequence need not be uniformly bounded. This is the exact point where the construction escapes the source paper's observation that a uniformly bounded $\mathcal{F}$ - π -basis contains an ordinary π -basis subsequence.

This result does not address the usual Frechet-filter π -property, nor does it answer whether the classical π -property implies an FDD.

Novelty Check

The local run indexes `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` were searched for 2508.21242, the source title, FDD, finite-dimensional decomposition, and close filter- π phrases; no prior packet for this target was found. A bounded external web search on June 28, 2026 using the exact title and phrases such as "every separable Banach space" "filter" "pi-property" "finite-rank projections" found the source arXiv page but no existing statement of this universal counterexample route. This is not a comprehensive MathSciNet/Zentralblatt search.

References

- [1] T. Kania and J. Swaczyna, *The π -property of a Banach space along a filter*, arXiv:2508.21242, 2025.
- [2] P. Enflo, *A counterexample to the approximation problem in Banach spaces*, Acta Mathematica 130 (1973), 309–317.